

Yugank Mishra

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Education

University of Texas at Dallas, Class of 2029

August 2025 - Present

Bachelor of Science, Major in Computer Science

Bachelor of Science, Major in Mathematics

- Member of CS², **Computing Scholars** (Computer Science Honors Program)

Honors and Awards

- Summer Platform for Undergraduate Research (SPUR) Poster Dept. of Computer Science Award
Winner, **Clark Summer Research Program**

August 2025

- National Merit Scholarship Recipient

March 2025

Work Experience

Clark Summer Research Program

May 2025 - August 2025

- Investigated gradient-based efficiency attacks on ML pipelines for **Automatic Speech Recognition** in Dr. Wei Yang's Lab for **10** weeks.
- Engineered a custom efficiency segmentation loss on a **CNN to Transformer** pipeline to generate attack vectors.
- Computed attack results on a Automatic Speech Recognition Pipeline processing AMI Corpus data, increasing **inference time** by **174%**, **memory usage** by **67%**, and **power consumption** by **10%**.

MI Lab, Research Assistant

September 2025 - September 2026

- Built a **ML powered, VR Hand pose** analysis tool, utilizing nearest neighbors with a tunable threshold for **custom gesture recognition in real time (1-2s)**.
- Synthesized a proprietary dataset of **2+ hours** of hand-pose telemetry & trained a robust hand pose **64** dimension encoder, computing embeddings in **milliseconds**.
- Submitted 2 papers to CHI '27.

Projects

Yope3D | [Github](#) | C++, Vulkan

October 2023 - Present

- Developed a **custom 3D game engine** from scratch with both **rasterization** and **ray-tracing** rendering, supporting 3D models, **physics**, audio, and scripting capabilities for all features.
- Enabled easy loading and usage of 3D models from **Blender**, with support for texturing, physics, etc, via extensive scripting support.
- Built an advanced **UI system** with **full FreeType support (custom fonts)**, **component design**, and easy integration with **2D/3D** game graphics for **overlays, menus**, and more.
- Implemented an multithreaded **physics system** capable of simulating **1000s** of **rigid-body** interactions with **constraint support** for game-ready physics (**doors, cars, ragdolls**).
- Created extensive documentation on GitHub, supplemented by [YouTube demos](#) and [meticulous devlogs](#) explaining the engineering.

ACM Projects | [Github](#) | React, Node.js, TypeScript, Supabase

Aug 2025 - Dec 2025

- Developed **SpinStack**, a full-stack music discovery and sharing platform with a **team of 5 developers**, after being selected as **1 of 55** participants, and presented to a **panel of 5 Industry judges**.

MNIST β -VAE Digit Drawer | [Github](#) | Python, Keras, Tensorflow, Numpy

November 2024

- Developed a neural network model that generates realistic images of handwritten digits (**β -VAE**); a foundational architecture for **diffusion models** in image generation.
- Controlled digit generation by varying **16 self-learned writing style parameters** varying across image generation, with an interactable showcase through a Colab Notebook.

LLM Mechanistic Interpretability Research | [Github](#) | [Paper](#) | Python

December 2025 - February 2026

- An exploratory investigation into **Typo Correction** by LLMs. Originally a submission to Google's MATS '25 Winter cohort.
- Using logit lensing, activation patching, and headwise ablation, the mechanistic interpretability analysis shows the tested models decide to induct typos in very late model layers.

Technical Skills

- **Languages:** Python, Java, C/C++, C#, TypeScript
- **Frameworks:** Unity, React, Tensorflow, PyTorch, LWJGL, Keras, OpenGL, Vulkan